

CSIP5102 (PART 1- Business intelligence)

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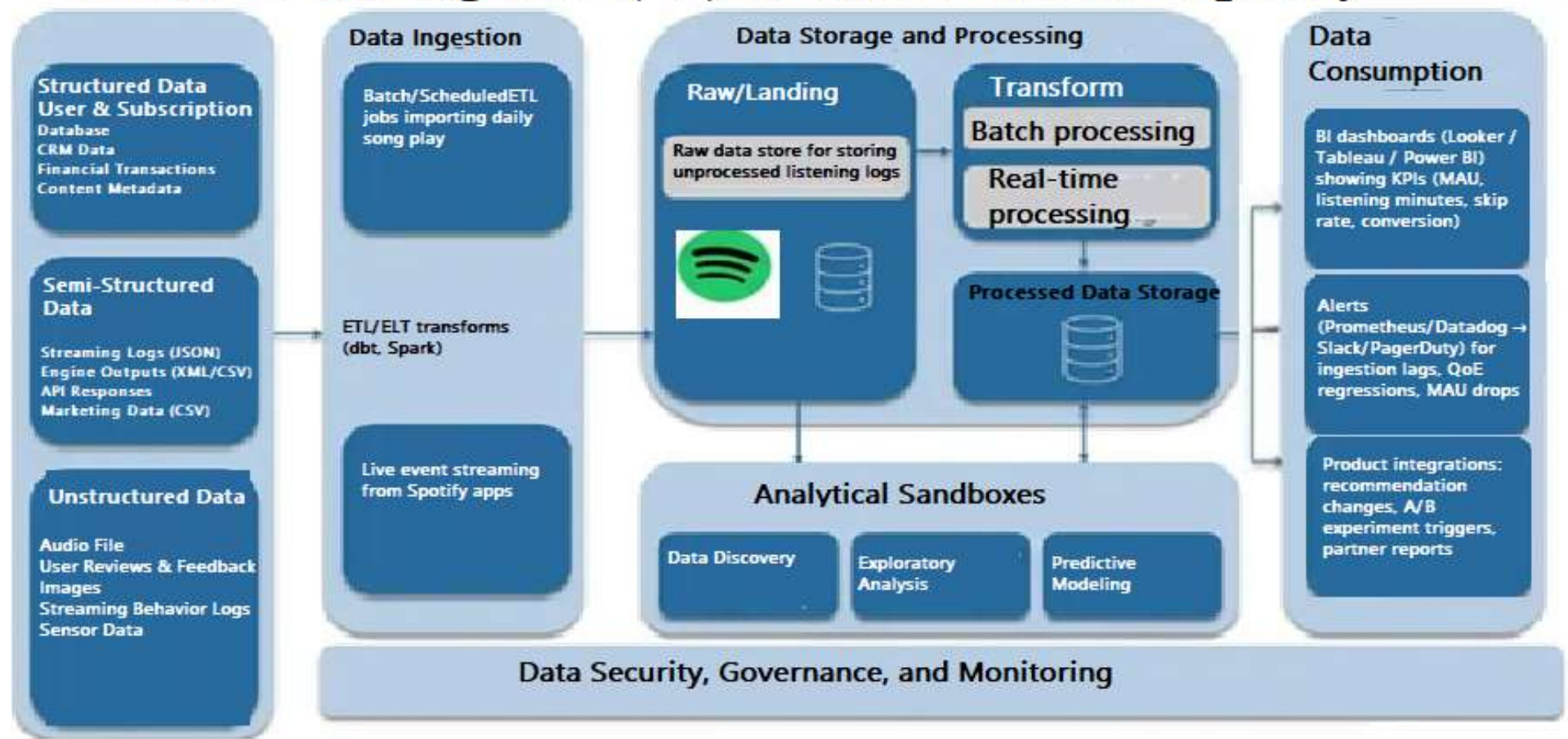
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Task 2 BI Architecture

Question 1 BI architecture design for Spotify

The below diagram is shows conceptual business intelligence (BI) architecture for Spotify:

Business intelligence (BI) architecture for Spotify



This architecture helps firm to ensure an effective and seamless flow of data from several structured and unstructured sources into centralized cloud based repositories in which data is processed, evaluated and visualised using dashboard for strategic decision making(Zhu et al., 2023).

Question 2: Internal and external data sources

Spotify's BI system integrate external and internal data sources using APIs as well as streaming technologies which support online transactions processing (OLTP) and analytics Suman, 2024).

Internal data sources

- **User interaction data (OLTP)**
 - Song streams, clicks, playlists, likes, skips and download capture through Spotify application server (Liang et al., 2025)
 - This forms real time operational data utilised for performance dashboard
- **Subscription and billing system**
 - The data and information from Spotify premium subscription, ad impression and payment get way Suman, 2024)
 - To support revenues forecasting and analysis (Ali and Kim, 2022)
- **CRM system**
 - It helps to contain customer profiles, churn analysis and engagement data
 - It enabled segmentation as well as targeted marketing campaigns
- **Operational and HR system**
 - It includes employees performance data and information, project durations and resource allocation
 - It is used for internal performance monitoring as well as workforce planning (Akter, and Kudapa, 2024)

External data sources

- **Social medial platform**
 - It helps to gather and collection users engagement and sentiment data for trend analysis and customer feedback monitoring (Liang et al., 2025)
- **Third party APIs**
 - It helps to provide competitors insights, industry benchmarks and regional performance comparison (Ali and Kim, 2022)
- **Advertising and parkers platform**
 - The data and information from external ad partners including podcast sponsorship, Meta ads and Google ads (Akter, and Kudapa, 2024)
- **Open source APIs and public data**
 - Data obtained from government, demographic datasets or regional music chart to contextualised customer growth trends

Therefore, together these sources ensured comprehensive visibility across firm's ecosystem and also aid them to support insight into development, monetization and enragement rate.

Question 3: considered unstructured data

Yes, Spotify considered unstructured data which is vital for data sources. According to Liang et al., (2025) stated that unstructured data (including logs, image, text and user feedback) form main component of firm's analytic ecosystem.

Reasons

- **Improve customer personalisation and experiences**
 - Spotify utilises unstructured behavioural and audio data in order to power its recommendation engines ("Discover Weekly," "Daily Mix")
 - ML(Machine learning) and NLP evaluate listening pattern, emotional tones and lyrics of song to recommend content (Akter, and Kudapa, 2024)
- **Sentiment and market insight**
 - Text mining form social media as well as customer review aids to determine customer segment and brand perceptions
- **Ad targeting and content creation**
 - Contextual and audio data from user session optimize ad placement, ehanchcing monetization efficiency (Ali and Kim, 2022)

- **Product development**

- Insight form unstructured podcast transcripts or user generated playlists helps to inform production innovation (Zhu et al., 2023)

Therefore from finding it has been found that unstructured data is processed implementing data lakes (AWS S3, Google Cloud Storage) ingrate with ML/AI pipelines to drive insight which structured data alone cannot reveal

Question 4: Aligned with the business needs and industry trend

Spotify’s BI architecture effectively aligned with the business needs and industry trend mainly those showed in McKinsey’s article “How to Build a Data Architecture to Drive Innovation—Today and Tomorrow.”

McKinsey Shift	Application in Spotify BI Architecture
From data silos to data products	The firm Spotify develop modular; domain oriented data product, podcast data product (listener engagement), Music Data Product (streaming stats) as well as ads data product. Thus each is owned by cross functional team and also ensuring reusability as well as consistency (McKinsey, 2024)
From pipelines to data ecosystems	The firm Spotify integration into cloud storage, APIs and partners data trend into unified ecosystem facilities them to real time data sharing across operations, finance and marketing(Ali and Kim, 2022)
From limited insight to AI-powered intelligence	AI and predictive modelling forecast churn, optimised pricing as well as helps them to detect fraud. Machine learning enriches BI dashboard with real time suggestion as well as automated insights
batch to real-time data flow	The technologies including Kafka and Google Dataflow has enabled streaming analytics and allowing executive to monitoring regular revenues as well as user trends dynamically (Zhu et al., 2023)

rigid architecture to agile, cloud-native systems	Spotify data Lakehouse architecture includes data lakes and data warehouse that enabled agile and scalable analytics with minimal latency
manual governance to automated governance	Spotify implements Data Catalogs and Metadata Management for automatic lineage monitoring access control and compliance and ensuring trust within BI data

Business needs and trend driving BI design

- **Cloud computing:** Spotify BI run on cloud computing platform (AWS, Google Cloud) in order to accomplish cost efficiency, scalability and global access for analytical time(Zhu et al., 2023)
- **Data lakehouse model:** Integrating structured (Revenues, customers) and unstructured (audio, feedback) data Spotify develop unified analytical platform which ensured flexiaibltiy and fast data exploration
- **Agile and continuously improvement:** The BI system effectively support agile decision making which allowing firm Spotify to effectively executive monitoring performance KPIs in real time and respond to market shift(Ali and Kim, 2022)
- **Innovation and comeptive advantage:** The BI design Spotify to innovate efficient through determining underperforming region, high growth and awn audience segment that improve its leadership within digital much streaming

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